

Lecture 22: Free Oscillations

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Michaelmas term 2024

Outline and Motivation

Today we begin our study of the Earth's **free oscillations**, also known as its normal modes. After a large earthquake, the whole Earth "rings like a bell" at a set of discrete frequencies.

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- We will analyse the resulting eigenvalue problem for a non-rotating Earth to find the properties of the normal modes.
- We will show how the forced response to an earthquake has **resonances** at the Earth's eigenfrequencies.
- Observing these frequencies provides the main constraint on the Earth's density structure.

A More Abstract Formulation

The full linearised equations of motion are complex. To simplify, we rewrite them in a **weak form**.

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Operator Notation

The resulting weak form can be written very concisely using a notation similar to the bra-ket formalism:

$$\langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | W | \dot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle = \langle \mathbf{w} | \mathbf{f} \rangle$$

Properties of the Operators

The operators P , W , and H represent the physical terms in the equation and have important symmetry properties.

- **P (Kinetic Energy):** Represents inertia ($Pu = \rho u$). It is **self-adjoint** ($\langle \mathbf{w}|P|\mathbf{u} \rangle = \langle \mathbf{u}|P|\mathbf{w} \rangle^*$) and defines a mass-weighted inner product.

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- **H (Potential Energy):** Represents elastic, gravitational, and centrifugal potential energy. It is **self-adjoint**.

These properties are crucial for understanding the nature of the free oscillations.

The Non-Rotating Case: An Eigenvalue Problem

To begin, we neglect rotation ($W = 0$) and external forcing ($f = 0$).
The equation becomes:

$$\langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle = 0$$

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This is a generalised eigenvalue problem for the eigenfrequencies ω and eigenfunctions (mode shapes) $\mathbf{s}$.

Properties of the Normal Modes (1/3): Reality

Because P and H are self-adjoint, we can prove key properties of the solutions.

Eigenfrequencies are Real

Sketch Proof:

- Let $\mathbf{s}$ be an eigenfunction with eigenfrequency ω . The eigenvalue equation is $\langle \mathbf{s} | H | \mathbf{s} \rangle = \omega^2 \langle \mathbf{s} | P | \mathbf{s} \rangle$.

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- Therefore, $\omega^2 = \frac{\text{real number}}{\text{positive real number}}$ must be a real number.

Properties of the Normal Modes (2/3): Orthogonality Setup

Eigenfunctions are Orthogonal

Sketch Proof (Part 1):

- Let $\mathbf{s}_1$ and $\mathbf{s}_2$ be two modes with distinct eigenfrequencies, $\omega_1^2 \neq \omega_2^2$.

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- Our goal is to show that $\langle \mathbf{s}_2 | P | \mathbf{s}_1 \rangle = 0$.

Properties of the Normal Modes (3/3): Orthogonality Proof

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Sketch Proof (Part 2):

- We use the self-adjoint property of H : $\langle \mathbf{s}_2 | H | \mathbf{s}_1 \rangle = \langle \mathbf{s}_1 | H | \mathbf{s}_2 \rangle^*$.

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- Taking the complex conjugate of Eq. (2) and using the self-adjointness of P gives:

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- Now the left hand sides are equal, so we can equate the right hand sides of this and Eq. (1):

$$\omega_1^2 \langle \mathbf{s}_2 | P | \mathbf{s}_1 \rangle = \omega_2^2 \langle \mathbf{s}_2 | P | \mathbf{s}_1 \rangle$$

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Forced Response and Resonances

Using the complete set of orthonormal eigenfunctions $|km\rangle$, we can solve the forced problem.

The Frequency-Domain Solution

The solution for the displacement in the frequency domain is found by an eigenfunction expansion:

$$\tilde{\mathbf{u}}(\omega) = \sum_{km} \frac{\langle km | \tilde{\mathbf{f}} \rangle}{\omega_k^2 - \omega^2} |km\rangle$$

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- After a large earthquake, the Fourier transform of a long seismogram shows a series of sharp peaks.
- The locations of these peaks are the Earth's normal mode

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Elastic vs. Gravitational Energy

Imagine uniformly compressing a planet.

- Elastic energy increases (resists compression).
- Gravitational binding energy decreases (becomes more negative).

The planet is stable only if the elastic energy increase outweighs the gravitational energy decrease for all possible deformations.

The Effect of Rotation

Including rotation re-introduces the Coriolis term into the eigenvalue problem:

Non-Linear Eigenvalue Problem

$$-\omega^2 \langle \mathbf{w} | P | \mathbf{s} \rangle + i\omega \langle \mathbf{w} | W | \mathbf{s} \rangle + \langle \mathbf{w} | H | \mathbf{s} \rangle = 0$$

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- The mathematics is more complex, and the eigenfunctions are no longer orthogonal in the simple sense.
- However, the fundamental result remains: the forced response still exhibits resonances at a discrete set of eigenfrequencies.

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- The forced response to an earthquake can be found using an **eigenfunction expansion**, which shows **resonances** at the eigenfrequencies.
- The **stability** of the Earth depends on a balance between stabilising elastic forces and destabilising gravitational forces.